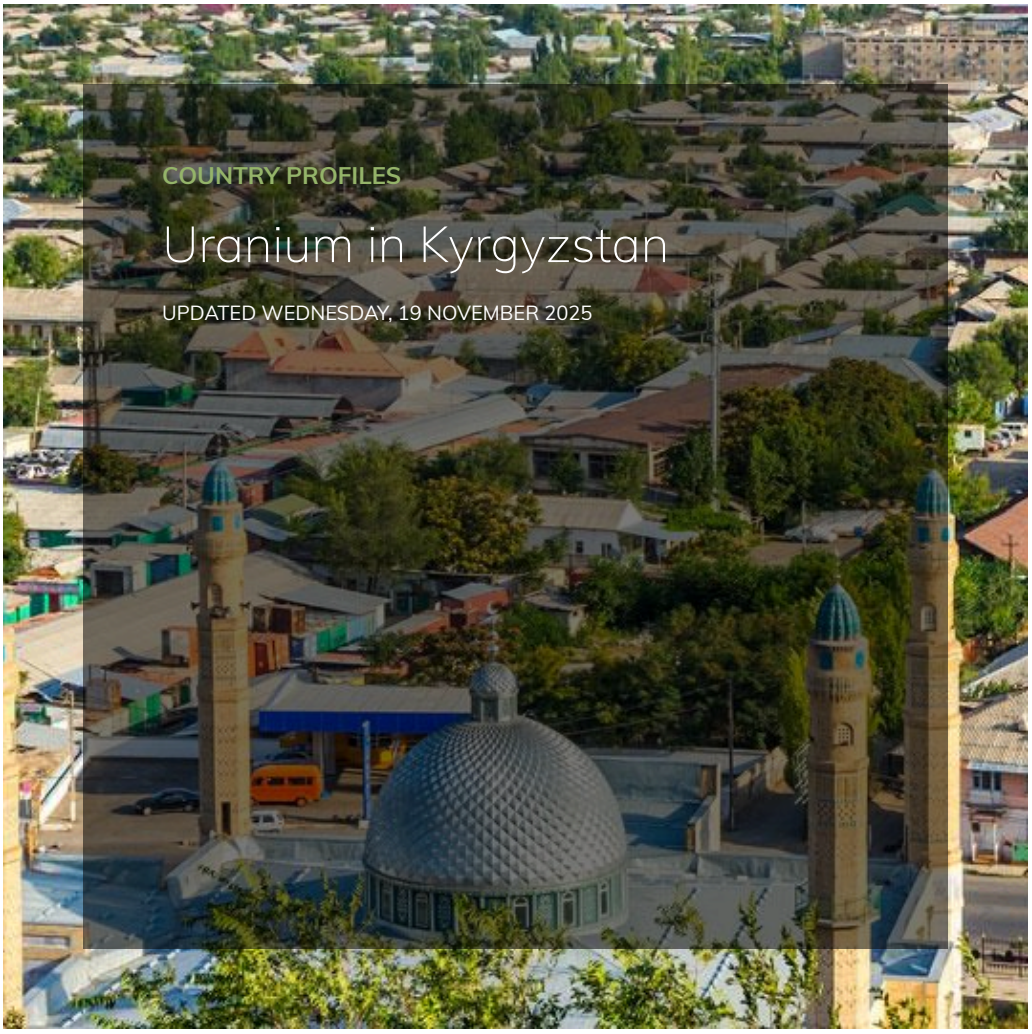

COUNTRY PROFILES

Uranium in Kyrgyzstan

UPDATED WEDNESDAY, 19 NOVEMBER 2025

- Kyrgyzstan has some uranium deposits and has supplied Russia in the past.
- It is the focus of remediation work on legacy uranium tailings in central Asia.
- In June 2024 the Kyrgyz parliament voted to lift the ban on uranium mining and exploration that had been in place since May 2019.

Generation mix: hydro 12.0 TWh (87%); coal 1.6 TWh (11%); natural gas 0.2 TWh.

Privacy - Terms

Import/export balance: 3.4 TWh net import (3.5 TWh imports; 0.1 TWh exports)

Total consumption: 14.1 TWh

Per capita consumption: c. 2000 kWh in 2023.

Source: International Energy Agency and The World Bank. Data for year 2023.

Having been annexed by Russia in 1876, Kyrgyzstan became a Soviet republic in 1936 and achieved independence in 1991.

The country has a history of uranium mining, but reports no uranium resources today (its last report in the IAEA 'Red Book' was 2002). The State Agency on Geology and Mineral Resources of the Kyrgyz Republic is the public agency that implements policy in the field of subsoil use and the development of the mining industry. The country's Mineral Resources Act was amended in 2014.

The Mailuu-Suu district, in the Jalal-Abad Province in western Kyrgyzstan was a significant uranium mining area in the Soviet era to 1989 where more than 9000 tonnes of uranium was produced between 1946 and 1967 from carbonate ore by the Western Mining and Chemical Combine. The Kara Balta Mining Combine was set up in the 1950s to mine and treat this ore in the north, near Bishkek. After Kyrgyz uranium mining ceased, in 1997 it became a joint stock company, Kara Baltinski Ore Mining & Processing Combine, and in 2007 the 72% state equity was purchased by the Renova Group – see below.

Meanwhile in the Soviet era, the Kyrgyzski Mining Combine was involved in mining in southern Kazakhstan.

Electricity generation in Kyrgyzstan is mainly from hydro (over 90%). The country has no nuclear power, but in January 2022 signed a memorandum of cooperation with Russia's Rosatom on the construction of small nuclear power plants based on the RITM-200N.

In January 2023 Rosatom announced that it was exploring the possibility of constructing a two-unit RITM-200 in the country. The high seismic activity in Kyrgyzstan has been cited as a challenge for site location.

In June 2023 Russia's Rosatom signed a memorandum of cooperation with Kyrgyzstan on healthcare applications of nuclear technologies.

Uranium projects and exploration

In May 2019 the Krygyz Supreme Council voted to ban uranium mining and exploration, except for uranium recovery from remediation work on legacy tailings. In December 2019 the President signed into law a ban on geological subsoil surveys related to the prospecting, exploration and development of uranium and thorium deposits in the country. The law prohibited import of uranium in any form, effectively terminating the Kara Balta mill (see section below). In June 2024 the Kyrgyz Parliament voted to lift the ban.

Azarga Uranium (formerly Powertech), registered in Canada, through its 70% subsidiary UrAsia has the Kyzyl Ompul project 125 km east of Bishkek, a uranium and rare earths lease. The main deposit is Kok Moinok, a hydrothermal deposit discovered in 1953 containing 2900 tU inferred resources (NI 43-101). The Tash Bulak and Backe placer deposits were part of the project, evidently with more resources. In August 2017 Azarga announced that Mining Investment Company Alliance had taken an option to acquire the Kyzyl Ompul project for $7.6 million. This was replaced by the similar earn-in agreement with Central Asian Uranium Co Ltd in April 2018, for $5.9 million cash and obliging it to spend $1.5 million on exploration and development. UrAsia will retain a smelter royalty of up to $5 million. In May 2019 Azarga suspended activities on the project after the parliament voted to ban uranium mining and exploration, and pending clarification of its licence situation.

US-based IMC Invest Inc has a number of deposits including the Kamushanovskoye deposit, with uranium adsorbed onto peat, about 70 km northeast of Kara Balta on the Kazakh border and 48 km north of Bishkek. It proposed a satellite ISL recovery there by alkaline leaching with the loaded resin processed at Kara Balta, total 2300 tonnes over ten years. Measured and indicated resources are about 2500 tU (JORC compliant). It has also explored the Jetym black shale deposits several hundred kilometres southeast of Kara Balta, with 48,700 tU in 'prognostic resources'.

China's Hebai Mining in 2011 bought from Raisama Ltd the major interest in the Kashkasu project, west of the historic uranium mines,

with uranium mineralization spread over 2.6 km strike length.

Central Asia Mining Co in 2012 had a licence to exploit Kara Balta tailings for uranium and molybdenum.

JSC Kentor had a licence to explore for uranium and other minerals at Bashkol.

Monaro Mining NL based in Australia had eight exploration licences which were prospective for uranium. These projects included Aramsu, Utor, Naryn, Sumsar, Sogul, Djurasay, Hodjaakan, and Gavasai (the last few including base and precious metals). In 2014 the company name became AusROC Metals Ltd and it appeared to be no longer involved with uranium. It was liquidated in 2016.

A number of other companies including ARMZ's Uranium One have explored for uranium. Nimrodel had leases in the Mailuu-Suu area.

In 2012 the Kyrgyz State Agency for Geology and Mineral Resources issued 18 permits for development of uranium and rare earth metal (REM) deposits and 21 permits for exploration of uranium and REM.

Several of the licences had CIS (Commonwealth Independent States) defined resources. However, they did not conform to JORC or NI 43-101 standards and a significant amount of further work would be required before JORC-compliant resource statements could be reported.

Uranium mill

The Kara Balta mining plant (KBMP), a hydrometallurgical plant, was built in 1951 on the Chu River near Bishkek to treat uranium ore from two Kyrgyz deposits (Kadji-Sai and Kavak) to 1989 and four Kazakh deposits (including Stepnoye and Tsentral) to 2005, and at its peak contributed 20% of USSR's uranium production. It also treated molybdenum ores and gold ores, and produced other metals and barite. It continued toll-milling operations until 2015 and formally closed a year later (see below).

The **JSC Kara Baltinski** Ore Mining & Processing Combine (KMPC) processed and calcined uranium product from Kazakhstan's Zarechnoye joint venture at the Kara Balta mill. Uranium dioxide was apparently the product. The Kyrgyz government held a 0.66% share in that JV. It accepted a tender from UralPlatina Holdings, a subsidiary of Russian resources investment group, Renova, for its 72.28% stake in the Kara Baltinski company in March 2007. The

€1.8 million price was less than half the earlier asking price, but it ended a four-year shutdown. The company, then named the Kara Balta Mining Enterprise (KGRK), struck an agreement in October 2008 with the Kazakh-based Eurasian Development Bank (EDB) to provide $150 million to develop the mill and properly emplace 50 years of tailings accumulation at Kara Balta. At the time the tailings were thought to contain up to 11,000 tU. However, they are very heterogeneous, posing a major metallurgical challenge.

Kara Balta then contracted with Russian-Kazakh Zarechnoye JV and Kazatomprom in Kazakhstan for toll milling. The mill resumed uranium production in August 2007 and increased annual production from about 800 tonnes of uranium to 2574 tU in 2009, exceeding for the first time historic output levels.

Early in 2013 KMPC was considering embarking upon mining in Kazakhstan, possibly in joint venture with Kazatomprom, or directly purchasing feed and selling the product to Russia, rather than toll milling. The company also was considering mining the Sary-Jaz group of deposits at Berkutsky, Schastlivy and Muzbulak projects. The prospective Kamyshanovskoye and Serafimovskoye deposits were also mentioned, with priority for exploration at Kyzyl-Ompul (Azarga Uranium holds a 424 sq km lease there).

However, in October 2015 the government said that KGRK might cease production due to lack of supply of uranium feed. It was appealing to Renova to secure more work for the mill. In November 2015 the mill ceased operation, due to lack of toll processing orders from Kazakhstan, and in February 2016 the company decided to close the plant and lay off staff. There were calls to return the enterprise to state ownership.

Legacy tailings

Apart from at Kara Balta, there are several tailings storage facilities in the Kyrgyz Republic, and more stretching 300 km to the west in Uzbekistan and Tajikistan. Some of them, especially around and above Mailuu-Suu in Jalal-Abad oblast which was threatened by a landslide in 2003, are in poor condition as no rehabilitation was done to either tailings storage facilities, waste rock dumps or uranium mining sites after Soviet mining ceased. Mailuu-Suu (or Mayli-Suu) has 23 uranium tailings dams containing 1.9 million cubic metres of material arising from the operation of Zapadnyi (Western) Mining and Chemical Combine to 1968 on the hillside above the town of 20,000 people. It is highly polluted by

radionuclides and heavy metals, and high incidence of cancer is reported. It is 25 km upstream of the densely-populated Fergana Valley with the Syr Darya River. Tuyuk-Suu in the Min-Kush valley in the centre of the country contained 430,000 cubic metres of uranium tailings, and the clay cover had degraded. In 1996 the Ministry of Environment led a $0.5 million project to assess and monitor uranium legacy issues in the southern part of Jalal-Abad oblast. The government considers these two sites as the highest priority for remediation.

The Industrial Association Southern Combine for Polymetals (IA Yuzhpolimetall) was established on the basis of the Kyrgyz Mining Combine founded in the 1960s. The Combine mined and processed uranium ores from the Southern Kazakhstan region to 1991. Throughout the period of the Combine's activities, 55 million cubic metres of below-ore-grade material and barren rocks in 18 dumps covering 214 ha and with 600 TBq activity were accumulated. These were all some 80 km from populated areas. Four tailings dams containing a total of 35 million cubic metres of fine tailings were formed on its sites, covering 258 ha and with 3200 TBq of activity. Three of the tailings dams, up to 11 km from Min-Kush, were finished between 1960 and 1969, after which half a metre of loam was compacted on the surface.

High-level discussions with Russia in 2004 in connection with setting up the Zarechnoye JV in Kazakhstan addressed the issue of rehabilitation of Kyrgyz tailings storages and uranium mining areas, particularly tailings near Kadgy Sai on the shore of Lake Issyk Kul in the east, Min Kush in the Tien Shan mountains, and Mailuu-Suu. The projected cost was $8.8 million. VNIPIpromtekhnologii (VNIPIPT) and ARMZ had looked at investments in projects related to rehabilitation of territories affected by uranium mining enterprises, including Kyrgyzstan.

Over 2010-12 a World Bank-funded pilot project that recovered one precarious tailings dam near Mailuu-Suu and reburied it at a cost of $8.4 million. This led to an estimate of $50 million to deal with ten other high-risk tailings in the area, and the Kyrgyz government appealed to the EU for this. A total of 36 waste piles and mill tailings have now been partially remediated and cultivated, and several landslide-prone spots near tailings have been improved and re-engineered to reduce the likelihood of seismic impact. This has been with assistance from the IAEA and some international funding. However, many of these projects remain incomplete, and many mines in need of remediation are in poor condition due to lack of funding.

In June 2015 the European Bank for Reconstruction and Development (EBRD) set up a fund to deal with radioactive contaminated material resulting from Soviet-era uranium mining and processing in the Kyrgyz Republic, Tajikistan and Uzbekistan. It said that the Environmental Remediation Account for Central Asia "is being established at the request of the European Commission, which is providing an initial €8 million, with additional funding under consideration." The fund would be used to finance remediation of "high-priority sites." It added: "Many of the legacy sites are concentrated along the tributaries to the Syr Darya River, which runs through the densely populated Fergana Valley, the agricultural centre of the region, which is shared by the Kyrgyz Republic, Tajikistan and Uzbekistan."

In August 2016 the Kyrgyzstan government announced rehabilitation works to begin in mid-2017, in collaboration with Tajikistan and Russia as well as the IAEA, and funded by the EBRD. The EBRD signed framework agreements with the Kyrgyz Republic and Tajikistan in January 2017, and the Kyrgyz government ratified its agreement in August 2017. ARMZ's VNIPIPT has opened an office in the capital, with a view to prioritizing work at Min-Kush, Kaji-Sai and a site near Taboshar in Tajikistan. The EU has carried out the environmental assessment and feasibility study for Min-Kush (Zhumgal district of Naryn region) and Shekaftar (Chatkala district, Jalal Abad oblast) near Mailuu-Suu, and contributed €16.5 million. Under a contract awarded in 2017, a consortium of companies is conducting an integrated environmental impact assessment and feasibility study for the management and remediation of Mailuu-Suu. The consortium comprises: GeoConsult of Austria; Belgium's Nuclear Research Centre (SCK-CEN); Wismut and Wisutec of Germany; and, Facilia of Sweden. Wismut and Wisutec have experience of remediating former uranium mines in the German states of Saxony and Thuringia. EBRD preliminary estimates suggested that the two main Kyrgyz sites would cost about €30 million. The Kyrgyz minister of emergency situations said that urgent measures were needed for environmentally unstable uranium legacy sites in Min-Kush and Shekaftar. Operations would be guided by a strategic master plan defining priority sites, prepared under the leadership of the IAEA. Remediation work managing the legacy waste from the sites in Shekaftar and Min-Kush was completed at the end of March 2022.

In September 2022 a contract was signed with Rosatom for the remediation of the former Taboshar mine and processing site near the city of Istiklol in western Tajikistan. The work was completed ahead of schedule in October 2023.

Earlier in 2021 the IAEA estimated that an additional $47 million would be required to tackle the most urgent problems caused by radioactive and toxic waste at legacy uranium sites in Central Asia, including those in Kyrgyzstan.

Non-proliferation

Kyrgyzstan is a party to the NPT and the Comprehensive Nuclear Test Ban Treaty (CTBT), and signed an Additional Protocol agreement with the IAEA in 2007. It has also ratified the Central Asia Nuclear Weapon Free Zone treaty, with Kazakhstan and Uzbekistan.

Notes & references

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